

Annexure No.:

EXPERIMENT - 01

Aim : Introduction to Breadboard and digital Multimeter, Verify the circuit analysis (voltage and current)

Apparatus : 3 resistor, 9V battery, breadboard, connectors, LED, multimeter.

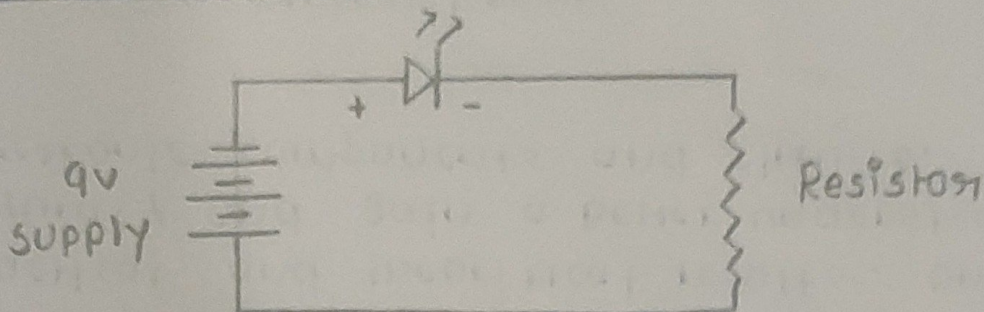
Theory :

A breadboard is a widely used tool in electronics for building and testing ~~etc~~ circuits without the need for soldering. It consists of a grid of holes with internal connections that allow components like resistors, wires, and ICs to be easily inserted and connected.

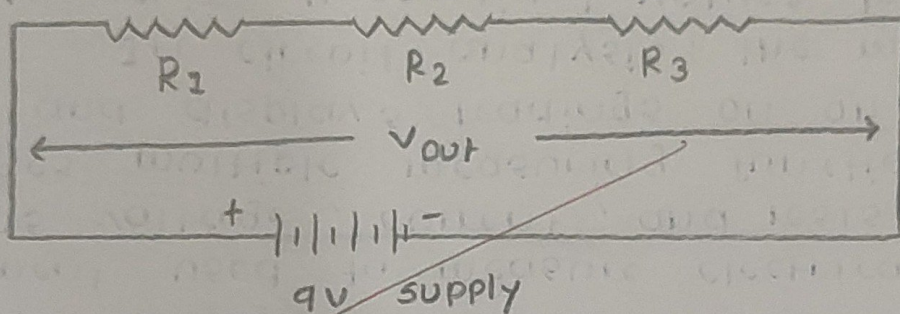
A digital multimeter (DMM) is an important instrument used to measure electrical quantities such as voltage, current, and resistance. It combines multiple measuring functions into one device and displays readings on an LCD screen.

In circuit analysis, the multimeter is used to verify theoretical values by measuring actual voltage and current in the circuit. This helps students understand practical behaviour and identify errors or deviations. By comparing practical and theoretical results, one can improve accuracy and gain a better understanding of electronic components and circuits.

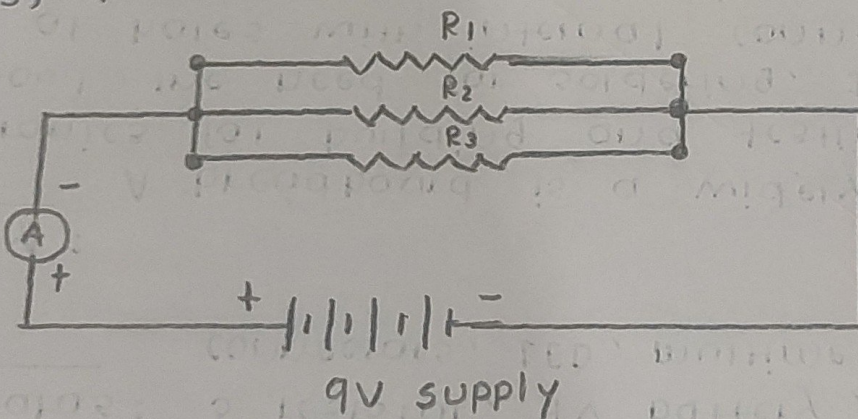
Circuit Diagram : (1) LED Blinking



(2) series connection :



(3) parallel connection :



Observation Table :

(1) Measurement of Resistor

Sr NO.	Theoretical value	Practical value
1	2.2 k Ω	2.17 k Ω
2	1 k Ω	0.95 k Ω
3	2.2 k Ω	2.17 k Ω

(2) Measurement of voltage

Sr NO.	Theoretical value	Practical value
1	2.23 V	3.67 k Ω V
2	1.22 V	0.96 k Ω V
3	2.25 V	3.68 V

(3) Measurement of current

Sr NO.	Theoretical value	Practical value
1	2.1 A	4.35 A
2	1.2 A	7.9 A
3	2.1 A	4.3 A

(4) Testing of Transistor

Transistor IC numbers	PNP/NPN
BC 547	NPN
BC 557	PNP

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Conclusion :

With the help of digital multimeter we can measure different electrical parameter with exact value of application.

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(2) MEASUREMENT OF CURRENT

2	3.32 A	3.32 A
3	1.55 A	0.48 A
1	8.53 A	3.81 A
2	MEASUREMENT OF VOLTAGE	MEASUREMENT OF VOLTAGE

(3) MEASUREMENT OF VOLTAGE

2	5.58 V	5.11 V
3	1.15 V	0.32 V
1	5.58 V	5.11 V
2	MEASUREMENT OF RESISTANCE	MEASUREMENT OF RESISTANCE

(4) MEASUREMENT OF RESISTANCE